

Plug-n-play e-knit with LED-based Sensor Positioning

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Abstract

The prototyping of electronic textiles (e-textiles) is challenging because of the need for complex and time-consuming wiring. This paper presents "plug-n-play e-knit," a prototyping platform, enabling large-area, reconfigurable on-textile sensor networks. Our method uses industrial knitting machines to embed wired power and data wiring throughout a garment with flexible magnetic connectors. This allows for repeated repositioning of on-textile sensors without damaging the textile. Additionally, LED-based CV approach can recognize the sensor's locations and IDs beforehand. We demonstrate that plug-n-play e-knit can provide fast e-textile prototyping over a 2700 cm² garment area.

CCS Concepts

• Human-centered computing → Interaction devices.

Keywords

E-textiles, e-knit, PME, wearable, plug-n-play, prototyping tool, smart clothing

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1 Introduction

The development of electronic textiles (e-textiles) for applications in activity recognition and healthcare often requires the repeated disassembly of integrated electronic devices to find the optimal position [2, 3]. For example, monitoring lower-limb rehabilitation demands the precise placement of sensors at a patient's specific injured joints and muscles [4]. Although e-textile prototyping tools aim to facilitate this proof-of-concept phase, they are frequently constrained by invasive connection methods such as stitching, welding, or adhesives. These techniques cause damage to the textile when sensing modules are moved, and the lack of methods for integrating large-area conductive pathways restricts the possible installation positions of sensors, preventing coverage over significant portions of the body [4].

To address these limitations, we introduce plug-n-play e-knit [1], a large-scale and reconfigurable e-textile prototyping tool. The system design is based on two main contributions: a machine-knitted communication and power supply network integrated into the textile, and a soft magnetic connector for rearranging modules. An industrial knitting machine directly fabricates a garment with a stretchable power and data bus at scale, which enables the free placement of modules across the textile. The connector, composed of Permanent Magnet Elastomer (PME), facilitates the repeated replacement of sensor modules without damaging the textile or impeding user motion. The experimental process can be further enhanced by an optional vision-based system that tracks on-module LEDs to automate the logging of each sensor's position and identifier. This integrated configuration provides a platform for repeatable on-textile performance evaluation and the development of applications like body motion tracking.

2 Demonstration

The system architecture of plug-n-play e-knit (Figure 1a) consists of three main parts: 1) an e-knit with several groups of exposed conductive lines, 2) a soft magnet connector which is compatible with a stretchable e-knit, and 3) sensing modules connected with the e-knit via the magnet connectors. The e-textile itself is fabricated using an industrial knitting machine that integrates conductive threads into

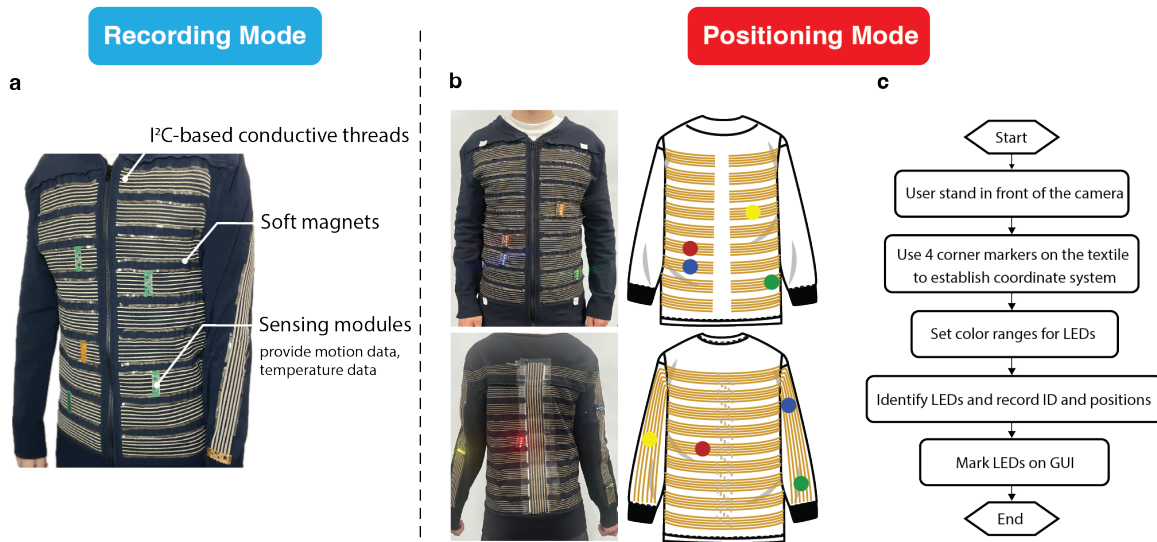


Figure 1: (a) Overview of plug-n-play e-knit comprising the I²C-based conductive threads, sensing module, soft magnet. (b) In positioning mode, the front and back of the garment with LEDs for detection. (c) The positioning workflow: establishing a coordinate system via corner markers, identifying LEDs by color, and mapping their positions to a GUI.

the garment, forming a six-line bus for 5V power (VCC/GND) and data transmission. To ensure signal integrity across the textile's large area, we employ a differential I²C protocol (SDA+/-, SCL+/-). Connectivity is provided by strips of stretchable PME affixed along the conductive bus, enabling a non-invasive, repositionable magnetic coupling with the sensor modules. Each lightweight, flexible module contains a low-power MCU, an IMU, and an I²C buffer. For automated spatial mapping, modules are equipped with programmable LEDs assigned a unique color as an identifier. An external camera and an OpenCV-based program detect the location and color of these LEDs, thereby registering the precise position of each specific module. This vision-based acquisition of spatial data is critical for developing diverse, context-aware e-textile applications.

Our demonstration will highlight the facile connectivity of the sensor network and the system's localization and identification capabilities. A user will wear the plug-n-play e-knit garment and stand before a camera. Our graphical user interface (GUI) will first identify the position of the e-textile by recognizing four white markers at its corners and the color of the conductive threads. We have prepared four sensor modules equipped with LEDs, each set to emit a unique color (red, blue, green, or yellow). These modules establish a stable connection to the power and communication network via their own magnets and the e-textile's PME, which illuminates the LEDs upon connection. This setup allows the user to freely place modules anywhere on the e-textile. The system then determines the position and ID of each module by its LED's location and color. Finally, the GUI will render a real-time display of the precise locations of the sensor modules (Figure 1b).

3 Conclusion

By combining a machine-knitted e-textile with repositionable magnetic connectors, our plug-n-play e-knit enables rapid, non-destructive

prototyping with sensor layouts. The integration of an LED-based vision system allows automated positioning of the sensor module. This work therefore establishes a robust methodology for creating large-area, reconfigurable e-textile networks.

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